

How the OA-1K TOLD Tool Was Verified

Verification & validation summary in plain language · Prepared September 11, 2026 · Tool version 1.9.16

What this document is

The OA-1K TOLD tool is an offline planning application that computes takeoff, landing, climb, cruise, and weight-and-balance numbers for the OA-1K. Every number it produces traces back to two authoritative sources: the aircraft's official flight-manual performance charts and the unit's signed weight-and-balance records. This report explains, in non-technical terms, how each function of the application was checked against those sources before being trusted — what was tested, how it was tested, and what the results were.

The short version: every calculation the tool performs was compared against the published source material it is supposed to reproduce, using several independent methods, and an automated 34-point test suite now re-verifies the whole application every time anything changes.

The core idea: the tool is a tracing, not an opinion

The flight manual delivers its performance data as printed charts — families of curves on graph paper. A pilot normally reads these by hand with a straightedge. The tool was built by digitizing those charts: each printed curve was traced point by point at high resolution (about 300 dots per inch, several thousand traced points per chart family), and the application simply reads the traced curves the same way a pilot would read the paper — no formulas were invented, no numbers estimated. Verification therefore means one thing: **proving the tracings faithfully match the printed ink, and that the software reads them correctly.**

Four independent methods were used, and every function was checked by at least two of them.

Method 1 — Reproducing the manual's own worked examples

Most flight-manual charts include a printed worked example: the publisher picks sample conditions, draws arrows through the chart, and prints the answer. These are, in effect, an answer key supplied by the aircraft manufacturer. The tool was given the exact same inputs, and its outputs were compared to the printed answers.

Function	Printed example result	Tool result	Agreement
Takeoff distance	Manual's worked chain	Matches	Within 0.2–2.7%
Accelerate-stop distance	Manual's worked chain	Matches	Within the same band
Landing distance	Manual's worked chain	Matches	Within the same band
Climb rate and gradient	625 ft/min, 6.0%, 101 kt	631 ft/min, 6.2%, 101 kt	Within 1%
Best-range cruise speed	Manual's printed answer	Exact match	Speed and fuel figures reproduced
Cruise torque / fuel flow	Manual's worked chain	Matches	Torque 0.6%, fuel flow 0.4%

A few percent is the same tolerance a careful human reader achieves with a straightedge on paper; the differences are within the width of the printed lines themselves.

Method 2 — Photographic overlay proof

Worked examples only test one point per chart. To verify the *entire* traced dataset, every traced curve was drawn back on top of a scan of the original printed page, in color, and inspected visually. If a tracing drifted anywhere along its length, it would be immediately visible as a colored line separating from the black printed one.

This overlay check was completed for all eight takeoff, landing, and accelerate-stop charts (every curve, end to end), both climb charts, and the engine-torque chart. The correction grids that adjust for aircraft weight, wind, and runway slope were verified the same way: the tool's internal correction tables were used to *redraw* the printed guide lines, and the redrawn lines were overlaid on the originals. In all cases the tracings sit directly on the printed ink.

For the engine-torque chart — which has no printed example to check against — 21 individual spot points across all seven curves were additionally verified with crosshair close-ups, and the software's readout was confirmed to match the traced chart within 9 foot-pounds (a fraction of one grid square) everywhere tested.

Method 3 — Independent cross-checks

Where the manual publishes the same information in two different forms, the two were checked against each other:

- The cruise charts and the separate specific-range charts describe the same physics independently. Values computed from one were compared against the other across the whole flight envelope, agreeing to within about 3.5% on average — confirming neither chart family was mistraced.
- Climb gradient is computed from climb rate and speed by a known formula; the computed gradients were checked against the manual's separately printed gradient lines and agree to better than 0.1%.
- The weight-and-balance calculator was run head-to-head against the unit's official Excel-based calculator using identical inputs. The results — zero-fuel weight, ramp weight, takeoff weight, center of gravity at every stage — matched the spreadsheet **exactly, to the decimal**, including the fuel burn-off sequence.

- The per-aircraft empty weights come from each tail number's signed weight-and-balance record (DD Form 365-F). Because those records were scanned documents, every value extracted was visually re-checked against an image of the original form, and where an aircraft had multiple records, the most recent was used.

Method 4 — Automated regression testing

Verifying something once is not enough if the software keeps evolving. A permanent automated test suite now opens the actual application in a browser, operates it like a user — selecting aircraft, entering fuel, planning missions — and checks 34 separate outcomes against known-correct answers: the flight-manual worked examples, the Excel-matched weight-and-balance checkpoints, the fuel-planning arithmetic, the safety limits, and the user-interface behaviors (over-limit warnings turning red, fields appearing and hiding correctly, the "Max Gas" fill respecting weight limits, and so on).

The suite runs after every change to the tool. All 34 checks pass on the current version. If any future edit breaks something that was previously verified, the suite catches it before the file is released.

Guardrails: what happens at the edge of the data

A planning tool is most dangerous when asked about conditions its data doesn't cover. This was tested deliberately: the tool was fed conditions hotter, higher, and heavier than the published charts. In every case it either refuses with a clear message or pins the answer to the edge of the published data *and says so on screen*. It was specifically confirmed that the tool never silently invents numbers beyond what the manual publishes.

The process catches real errors — including its own

Three findings during verification are worth highlighting, because they show the process has teeth:

1. **Errors found in the source document itself.** The digitization uncovered several drafting errors in the printed manual — for example, a worked example whose arrows were drawn entering the wrong curve. These were documented and reported rather than silently reproduced.
2. **An error found in the tool.** Verification of the climb charts revealed the tool was reading the wrong curve for maximum-weight climbs. It was corrected the same day, and a test was added so it can never silently recur.
3. **An error caught by the subject-matter expert.** During review, the squadron user spotted that a curve identity on the climb chart had been mislabeled during the fix above. The correction was verified against the chart geometry, applied, and locked in with two new automated tests. Human expert review remains part of the loop.

What this adds up to

- Every chart family in the tool has been verified against the printed source by at least two independent methods.
- The weight-and-balance function matches the unit's official calculator exactly.
- Aircraft data comes from signed records, individually re-checked against the originals.
- The tool refuses to guess outside its published data, visibly.
- A 34-point automated test suite re-proves all of the above on every future change.

The tool is a faithful, checkable reading of the official publications — nothing more, nothing less — and the evidence for that statement can be regenerated on demand.

A separate technical companion document, "OA-1K TOLD Tool — Data & Methodology," records the full engineering detail behind this summary for standards and evaluation personnel.